LECTURE 1
FUNDAMENTALS OF EV BATTERY PACK DESIGNS

An electric vehicle (EV) battery pack is a complete energy-storage assembly that supplies electrical power to the vehicle's electric motor and other electrical systems. It consists of multiple individual battery cells connected in series and parallel to achieve the required voltage, current, and energy capacity.

Battery pack design is important because it directly affects the vehicle's driving range, acceleration, charging time, safety, weight, cost, and overall performance. A well-designed battery pack must provide sufficient energy for driving while operating safely under different temperatures, loads, and road conditions.

BATTERY CELL, MODULE, AND PACK

An EV battery system is generally organized into three levels: cell, module, and pack.

- Battery cell: The smallest electrochemical unit that stores and delivers electrical energy.
- Battery module: A group of cells connected in series, parallel, or a combination of both and assembled into a single unit.
- Battery pack: A complete assembly of modules or directly connected cells, along with electrical connections, protection devices, cooling arrangements, and a battery management system.

The arrangement depends on the required voltage, energy capacity, power output, and vehicle design.

BATTERY CELL CONFIGURATION

An EV battery pack must satisfy several important requirements. The voltage rating must match the electric drive and power electronics. The energy capacity must be sufficient to provide the desired driving range, while the power capability must meet acceleration, hill-climbing, and regenerative-braking requirements.

The pack should also be lightweight, compact, safe, durable, and capable of operating over the expected temperature range. Effective thermal management is required to prevent overheating and maintain uniform cell temperatures. Electrical protection and a battery management system (BMS) are also necessary for reliable operation.

Battery cells are connected in series and parallel to obtain the required pack voltage and capacity. In a series connection, the positive terminal of one cell is connected to the negative terminal of the next cell. The voltages of the cells add, while the ampere-hour capacity remains approximately equal to that of one cell.

For N_S identical cells connected in series:

$$V_{pack} = N_s V_{cell}$$
$$Q_{pack} = Q_{cell}$$

where N_s is the number of cells in series, V_{cell} is the nominal voltage of one cell, and Q_{cell} is its capacity in ampere-hours (Ah).

For example, if 100 cells, each rated at 3.6 V, are connected in series:

$$V_{pack} = 100 \times 3.6 = 360V$$

Thus, the nominal pack voltage is 360 V.

In a parallel connection, all positive terminals are connected together, and all negative terminals are connected together. The voltage remains approximately equal to the voltage of one cell, while the capacities add.

For N_p identical cells connected in parallel:

$$V_{pack} = V_{cell}$$
$$Q_{pack} = N_p Q_{cell}$$

For example, if four cells, each having a capacity of 5 Ah, are connected in parallel:

$$Q_{pack} = 4 \times 5 = 20Ah$$

Thus, the parallel group has a capacity of 20 Ah.

In practical EV battery packs, series and parallel connections are combined to achieve the required voltage and capacity.

For identical cells arranged in N_s series groups and N_p parallel cells per group:

$$V_{pack} = N_s V_{cell}$$
$$Q_{pack} = N_p Q_{cell}$$

The nominal energy capacity is

$$E_{pack} = V_{pack} Q_{pack}$$

If voltage is measured in volts and capacity in ampere-hours, energy is obtained in watt-hours (Wh).

Example: Consider lithium-ion cells rated at 3.6 V and 5 Ah, arranged in a 100s4p configuration.

Here, 100 cells are connected in series in each string, and four such strings are connected in parallel.

$$V_{pack} = 100 \times 3.6 = 360\ V$$
$$Q_{pack} = 4 \times 5 = 20Ah$$
$$E_{pack} = 360 \times 20 = 7200wh = 7.2kWh$$

The pack contains $100 \times 4 = 400$ cells.

A battery pack is required to provide approximately 48 V and 30 Ah using cells rated at 3.7 V and 2.5 Ah. Determine the series count, parallel count, total number of cells, and nominal energy.

Series count: $N_s = \frac{48}{3,7} = 12.97 \approx 13$

Parallel count: $N_p = \frac{30}{2.5} = 12$

Total number of cells: $N_{total} = 13 \times 12 = 156$

Nominal voltage: $V_{pack} = 13 \times 3.7 = 48.1V$

Nominal energy: $E_{pack} = 48.1 * 30 = 1443Wh$

Answer: 13s12p, 156 cells, 48.1 V nominal, 30 Ah, and 1.443 kWh nominal energy.

An EV consumes 23 Wh/km and requires a range of 100 km. Estimate the ideal battery capacity at 48 V.

Required energy: $E = 23 \times 100 = 2300Wh$

Required capacity: $Q = \frac{2300}{48} = 47.92Ah$

Answer: The ideal calculated capacity is approximately 48 Ah. A practical battery should be sized after accounting for usable energy, losses, and design margin.

BATTERY PACK CONFIGURATION AND LAYOUT

Battery pack layout refers to the physical arrangement of cells, modules, electrical connections, and auxiliary components within the pack enclosure. Cells can be arranged in cylindrical, prismatic, or pouch formats. Cylindrical cells are enclosed in metal cans, prismatic cells have rigid rectangular enclosures, and pouch cells use flexible laminated packaging. The selected format affects space utilization, mechanical support, cooling arrangements, manufacturing, and maintenance. The cells must be positioned so that electrical connections are reliable and heat can be removed effectively.

Two common design approaches are:

- Module-based design: Cells are grouped into modules, and the modules are assembled into the battery pack.
- Cell-to-pack design: Cells are integrated more directly into the pack, reducing the need for separate module structures.

The choice depends on vehicle requirements, manufacturing processes, serviceability, structural design, and safety considerations.

BATTERY CELL TYPES AND PHYSICAL CONSTRUCTION

The presentation illustrates cylindrical cells and prismatic cells. Cell format refers to the physical shape and packaging of the electrochemical cell; it is different from the battery chemistry.

Cylindrical cells

Cylindrical cells have a round, tube-like metallic casing. They are commonly used in applications where numerous small cells are connected to form a larger battery pack.

Characteristics:

- Standardized cylindrical shape and dimensions for a given cell format.
- Convenient for automated manufacturing and assembly.
- Multiple cells can be arranged in series-parallel groups.
- The spaces between cells affect packaging efficiency and thermal design.
- Large numbers of cells may require extensive electrical interconnections and monitoring.

Prismatic cells

Prismatic cells have a rectangular enclosure. Their flat surfaces allow them to be arranged closely together in a battery module or pack.

Characteristics:

- Rectangular shape facilitates compact packaging.
- Fewer individual cells may be needed for a given pack capacity compared with smaller cylindrical cells.
- Mechanical support and compression may be important, depending on cell design.
- Heat distribution and temperature uniformity must be considered.
- Cell replacement and monitoring depend on the module and pack architecture.

The selection between cylindrical and prismatic cells depends on energy density, power requirements, cooling arrangements, mechanical constraints, manufacturing cost, reliability, and safety. Neither format is universally superior for every electric vehicle.